

The University of Azad Jammu &Kashmir

Muzaffarabad, AJK

Course Title

Software Design & Architecture

Course Code

SEC-532

Semester

3rd

Lab 03

Student Case Study

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Roll No

2025-SE-10

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Submission Date

November, 7, 2026

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Class 1: Student:

The Student class stores information about university students. Each student has a unique student ID and a name. A student can register for multiple courses.

Attributes:

studentID: String

name: String

Class 2: Course

The Course class stores information about courses offered by the university. It contains the course code, title, credit hours, and maximum seat capacity.

Attributes:

courseCode: String

title: String

creditHours: int

capacity: int

Class 3: Enrollment:

The Enrollment class represents the registration of a student in a particular course. It stores the registration date and current enrollment status. It is necessary because each registration has its own information.

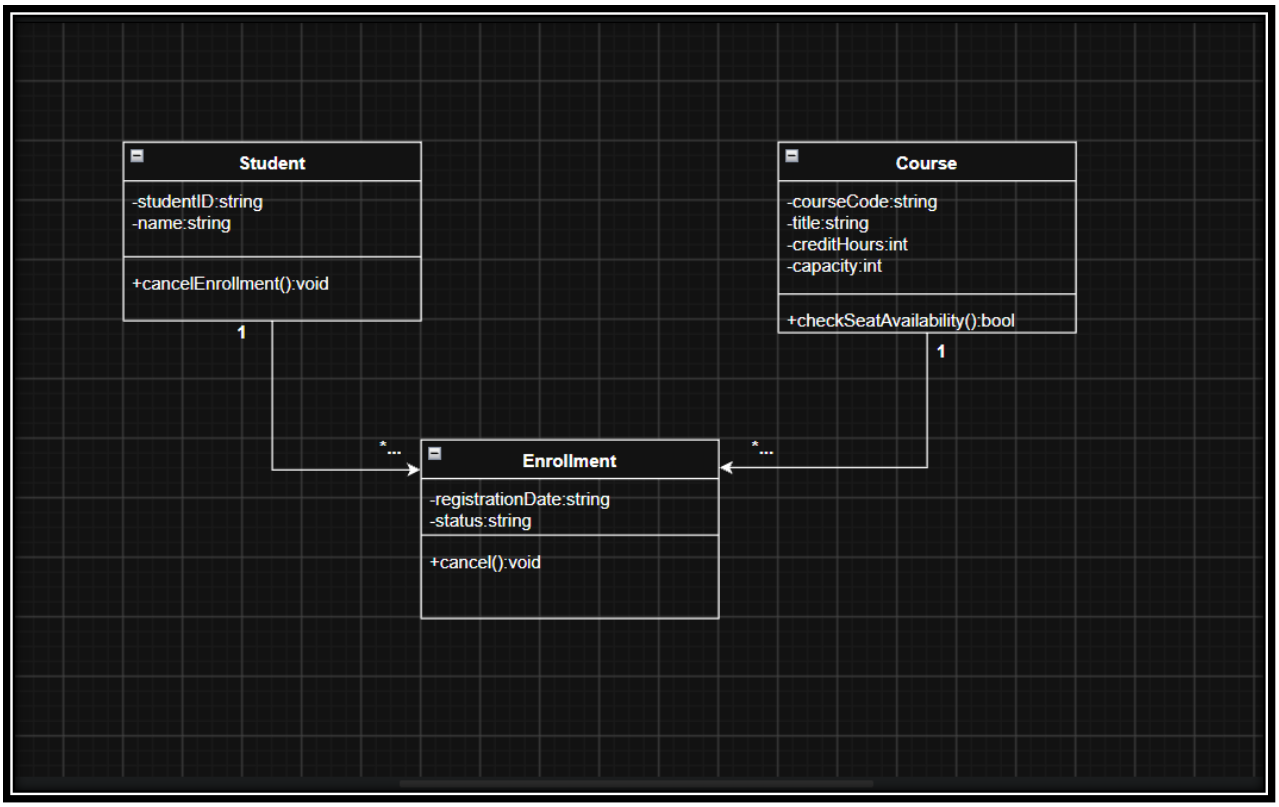
Attributes:

registrationDate: Date

status: String

2. Operations Explanation:

Class	Operation	Explanation
Student	cancelEnrollment():	Cancels an existing enrollment of a student.
Course	hasAvailableSeat():boolean	boolean Checks whether a course has an available seat
Enrollment	cancel():void	Changes the enrollment status to cancelled



- One Student can have zero or many Enrollments (0..*).
- One Course can have zero or many Enrollments (0..*).
- Each Enrollment belongs to exactly one Student (1).
- Each Enrollment refers to exactly one Course (1).

The multiplicity 0..* means that an object may have zero or many related objects. The multiplicity 1 means exactly one related object.

4. Answers to Lab Questions

Q10. Why is a registration record useful instead of linking a student directly to a course?

An enrollment record is useful because it stores additional information about a student's registration, such as the registration date and enrollment status. It also helps manage cancellation.

Q11. Can a student exist without being registered in a course?

Yes, a student can exist without registering in any course. The multiplicity 0..* between Student and Enrollment represents this situation.

Q12. Can a course exist even if no student has registered for it?

Yes, a course can be offered even if no student has registered. The multiplicity 0..* between Course and Enrollment represents this situation.

Q13. Which class should store the date and status of an individual registration? Why?

The Enrollment class should store the registration date and status because these details belong to a specific registration, not directly to the student or course.

Q14. Which operation could represent cancellation? Which could represent checking seat availability?

The cancelEnrollment() operation can represent cancellation. The hasAvailableSeat() operation can represent checking whether a course has an available seat.

Q15. Does your class diagram show the order in which messages are sent? Explain why or why not.

No, a class diagram does not show the order in which messages are sent. It represents the static structure of a system, including classes, attributes, operations, and relationships. A sequence diagram is used to show the order of messages.

5. Short Explanation (100–150 Words)

Explanation of Student Course Registration System:

The Student Course Registration System consists of three main classes: **Student**, **Course,** and **Enrollment**. The Student class stores the student ID and name. The Course class stores the course code, title, credit hours, and maximum seat capacity. The Enrollment class represents the registration of a student in a course and stores the registration date and enrollment status.

The Enrollment class is important because it maintains additional information about each registration and supports cancellation. A student can have zero or many enrollments, and a course can also have zero or many enrollments. However, each enrollment belongs to exactly one student and refers to exactly one course. The system also includes operations to cancel an enrollment and check seat availability. This class diagram represents the static structure of the registration system.